10th Grade
Term 1
C5 Practice Activity

Evaluated Criterion

C5: Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opportunity loss.

1 Minimax Regret Criterion

When probabilities are not available, a decision-maker can compare alternatives by considering the opportunity loss, or regret, associated with each possible state of nature. Regret measures the payoff that could have been obtained by choosing the best alternative for the state of nature that actually occurs, minus the payoff obtained from the selected alternative.

For a payoff table with alternatives in rows and states of nature in columns, let $P_{ay}(A_i, S_j)$ represent the payoff obtained when alternative A_i is selected and state of nature S_j occurs.

For each state of nature, first identify the best available payoff:

$$\begin{aligned} b(S_j) &= \max_i P_{ay}(A_i, S_j) \\ &= \max\left\{P_{ay}(A_1, S_j), P_{ay}(A_2, S_j), \ldots, P_{ay}(A_m, S_j)\right\}. \end{aligned}$$

The regret associated with alternative A_i under state S_j is then

$$r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j).$$

For each alternative, determine its maximum possible regret:

$$\begin{aligned} R(A_i) &= \max_j r(A_i, S_j) = \max_j \left\{b(S_j) - P_{ay}(A_i, S_j)\right\} \\ &= \max\left\{r(A_i, S_1), r(A_i, S_2), \ldots, r(A_i, S_n)\right\}. \end{aligned}$$

The Minimax Regret criterion selects the alternative whose maximum regret is as small as possible:

$$A^*_{\text{Minimax Regret}} = \arg\min_i R(A_i).$$

Thus, the procedure is:

> Best payoff in each state $\longrightarrow$ Regret table $\longrightarrow$ Maximum regret for each alternative
> Minimum of those maximum regrets

The selected alternative minimizes the largest opportunity loss that could result from the decision.

2 General 2x2 Problem

Consider the following payoff table:

Alternative	S_1	S_2
A_1	$P_{ay}(A_1, S_1)$	$P_{ay}(A_1, S_2)$
A_2	$P_{ay}(A_2, S_1)$	$P_{ay}(A_2, S_2)$

Construct the opportunity loss table and determine the decision using the Minimax Regret criterion.

2.1 Solution

Step 1: Symbols, values, and meanings.

Symbol	Value or identification	Meaning
A_1, A_2	Two available alternatives	Choices represented by the payoff-table rows.
S_1, S_2	Two possible states	Outcomes represented by the payoff-table columns.
$P_{ay}(A_i, S_j)$	Given symbolically	Payoff for choice A_i in state S_j, in the problem's monetary units.
$b(S_j)$	Calculated in Step 2	Best payoff in state S_j: the column maximum.
$r(A_i, S_j)$	Calculated in Step 3	Regret: the state's best payoff minus this choice's payoff.
$R(A_i)$	Calculated in Step 4	Largest regret for choice A_i: the row maximum.

No numerical data are supplied; this template uses symbolic calculations only.

Step 2: Find the best payoff in each state.

State	Symbolic calculations
S_1	$b(S_1) = \max\{P_{ay}(A_1, S_1), P_{ay}(A_2, S_1)\}$
S_2	$b(S_2) = \max\{P_{ay}(A_1, S_2), P_{ay}(A_2, S_2)\}$

Step 3: Calculate opportunity losses. Subtract each payoff from its own column's best payoff. The result of each subtraction is the regret identified on the second line.

Alternative	S_1	S_2
A_1	$b(S_1) - P_{ay}(A_1, S_1)$	$b(S_2) - P_{ay}(A_1, S_2)$
	$= r(A_1, S_1)$	$= r(A_1, S_2)$
A_2	$b(S_1) - P_{ay}(A_2, S_1)$	$b(S_2) - P_{ay}(A_2, S_2)$
	$= r(A_2, S_1)$	$= r(A_2, S_2)$

Step 4: Find the maximum regret for each alternative.

Alternative	Symbolic calculations
A_1	$R(A_1) = \max\{r(A_1, S_1), r(A_1, S_2)\}$
A_2	$R(A_2) = \max\{r(A_2, S_1), r(A_2, S_2)\}$

Step 5: Apply the Minimax Regret criterion.

$$R^* = \min\{R(A_1), R(A_2)\}, \qquad A^*_{\text{Minimax Regret}} = \arg\min_i R(A_i).$$

Choose A_1 if $R(A_1) < R(A_2)$, choose A_2 if $R(A_2) < R(A_1)$, and recommend both alternatives if $R(A_1) = R(A_2)$.

Step 6: Final symbolic opportunity loss table.

Alternative	S_1	S_2	Maximum regret
A_1	$r(A_1, S_1)$	$r(A_1, S_2)$	$R(A_1)$
A_2	$r(A_2, S_1)$	$r(A_2, S_2)$	$R(A_2)$

3 Contextualized 2x2 Problem - Coffee shop

A small coffee shop is considering whether to introduce a new dessert line or keep its current menu. The manager knows that the level of customer traffic during the promotional weekend is uncertain and has no reliable information about the probability of high or low traffic.

What decision should the manager make using the Minimax Regret criterion?

Alternative	High traffic (S_1)	Low traffic (S_2)
Launch dessert line (A_1)	42	8
Keep current menu (A_2)	28	20

All payoffs are expressed in thousands of dollars.

Determine the decision using the Minimax Regret criterion.

3.1 Solution

Step 1: Symbols, values, and meanings.

Symbol	Value or identification	Meaning
A_1	Launch dessert line	Alternative chosen before the state is known.
A_2	Keep current menu	Alternative chosen before the state is known.
S_1	High traffic	Possible state outside the decision-maker's control.
S_2	Low traffic	Possible state outside the decision-maker's control.
$P_{ay}(A_1, S_j)$	42, 8 (thousand dollars)	Payoffs for Launch dessert line, in S_1, S_2 order.
$P_{ay}(A_2, S_j)$	28, 20 (thousand dollars)	Payoffs for Keep current menu, in S_1, S_2 order.
$b(S_j)$	Calculated in Step 2	Best available payoff in state S_j (column maximum).
$r(A_i, S_j)$	Calculated in Step 3	Opportunity loss: the best payoff in this state minus this alternative's payoff.
$R(A_i)$	Calculated in Step 4	Maximum regret for alternative A_i (row maximum).

All payoff and regret amounts below are in thousands of dollars.

Step 2: Find the best payoff in each state.

Symbolic calculations

State	Comparison
S_1	$\max\{P_{ay}(A_1, S_1), P_{ay}(A_2, S_1)\} = b(S_1)$
S_2	$\max\{P_{ay}(A_1, S_2), P_{ay}(A_2, S_2)\} = b(S_2)$

Numerical calculations

State	Comparison
S_1	$\max\{42, 28\} = 42$
S_2	$\max\{8, 20\} = 20$

Step 3: Calculate opportunity losses.

Symbolic calculations

Alternative	S_1	S_2
A_1	$b(S_1) - P_{ay}(A_1, S_1) = r(A_1, S_1)$	$b(S_2) - P_{ay}(A_1, S_2) = r(A_1, S_2)$
A_2	$b(S_1) - P_{ay}(A_2, S_1) = r(A_2, S_1)$	$b(S_2) - P_{ay}(A_2, S_2) = r(A_2, S_2)$

Numerical calculations

Alternative	S_1	S_2
A_1	$42 - 42 = 0$	$20 - 8 = 12$
A_2	$42 - 28 = 14$	$20 - 20 = 0$

Step 4: Find the maximum regret for each alternative.

Symbolic calculations

Alternative	Row maximum
A_1	$\max\{r(A_1, S_1), r(A_1, S_2)\} = R(A_1)$
A_2	$\max\{r(A_2, S_1), r(A_2, S_2)\} = R(A_2)$

Numerical calculations

Alternative	Row maximum
A_1	$\max\{0, 12\} = 12$
A_2	$\max\{14, 0\} = 14$

Step 5: Apply the Minimax Regret criterion.

Symbolic calculations

Quantity	Comparison
R^*	$\min\{R(A_1), R(A_2)\} = R^*$
A^*	$\arg\min_i R(A_i) = A^*_{\text{Minimax Regret}}$

Numerical calculations

Quantity	Comparison
R^*	$\min\{12, 14\} = 12$
A^*	$R(A_1) = R^* = 12 = A_1$

Launch the dessert line (A_1): it minimizes maximum regret at \$12,000, compared with \$14,000 for keeping the current menu. There is no tie.

Step 6: Final numerical opportunity loss table (thousand dollars).

Alternative	S_1	S_2	Maximum regret
A_1: Launch dessert line	0	12	**12** (recommended)
A_2: Keep current menu	14	0	14

4 Contextualized 2x2 Problem - Clothing store

A local clothing store is deciding whether to launch a new advertising campaign before the holiday season or continue with its current marketing strategy. The manager knows that customer response to the campaign is uncertain and has no reliable information about the probability of strong or weak sales.

What decision should the manager make using the Minimax Regret criterion?

Alternative	Strong sales (S_1)	Weak sales (S_2)
Launch new campaign (A_1)	55	5
Keep current strategy (A_2)	35	22

All payoffs are expressed in thousands of dollars.

Determine the decision using the Minimax Regret criterion.

4.1 Solution

Step 1: Symbols, values, and meanings.

Symbol	Value or identification	Meaning
A_1	Launch new campaign	Alternative chosen before the state is known.
A_2	Keep current strategy	Alternative chosen before the state is known.
S_1	Strong sales	Possible state outside the decision-maker's control.
S_2	Weak sales	Possible state outside the decision-maker's control.
$P_{ay}(A_1, S_j)$	55, 5 (thousand dollars)	Payoffs for Launch new campaign, in S_1, S_2 order.
$P_{ay}(A_2, S_j)$	35, 22 (thousand dollars)	Payoffs for Keep current strategy, in S_1, S_2 order.
$b(S_j)$	Calculated in Step 2	Best available payoff in state S_j (column maximum).
$r(A_i, S_j)$	Calculated in Step 3	Opportunity loss: the best payoff in this state minus this alternative's payoff.
$R(A_i)$	Calculated in Step 4	Maximum regret for alternative A_i (row maximum).

All payoff and regret amounts below are in thousands of dollars.

Step 2: Find the best payoff in each state.

Symbolic calculations

State	Comparison
S_1	$\max\{P_{ay}(A_1,S_1), P_{ay}(A_2,S_1)\} = b(S_1)$
S_2	$\max\{P_{ay}(A_1,S_2), P_{ay}(A_2,S_2)\} = b(S_2)$

Numerical calculations

State	Comparison
S_1	$\max\{55, 35\} = 55$
S_2	$\max\{5, 22\} = 22$

Step 3: Calculate opportunity losses.

Symbolic calculations

Alternative	S_1	S_2
A_1	$b(S_1) - P_{ay}(A_1,S_1) = r(A_1,S_1)$	$b(S_2) - P_{ay}(A_1,S_2) = r(A_1,S_2)$
A_2	$b(S_1) - P_{ay}(A_2,S_1) = r(A_2,S_1)$	$b(S_2) - P_{ay}(A_2,S_2) = r(A_2,S_2)$

Numerical calculations

Alternative	S_1	S_2
A_1	$55 - 55 = 0$	$22 - 5 = 17$
A_2	$55 - 35 = 20$	$22 - 22 = 0$

Step 4: Find the maximum regret for each alternative.

Symbolic calculations

Alternative	Row maximum
A_1	$\max\{r(A_1,S_1), r(A_1,S_2)\} = R(A_1)$
A_2	$\max\{r(A_2,S_1), r(A_2,S_2)\} = R(A_2)$

Numerical calculations

Alternative	Row maximum
A_1	$\max\{0, 17\} = 17$
A_2	$\max\{20, 0\} = 20$

Step 5: Apply the Minimax Regret criterion.

Symbolic calculations

Quantity	Comparison
R^*	$\min\{R(A_1), R(A_2)\} = R^*$
A^*	$\arg\min_i R(A_i) = A^*_{\text{Minimax Regret}}$

Numerical calculations

Quantity	Comparison
R^*	$\min\{17, 20\} = 17$
A^*	$R(A_1) = R^* = 17 = A_1$

Launch the new campaign (A_1): it minimizes maximum regret at \$17,000, compared with \$20,000 for keeping the current strategy. There is no tie.

Step 6: Final numerical opportunity loss table (thousand dollars).

Alternative	S_1	S_2	Maximum regret
A_1: Launch new campaign	0	17	**17** (recommended)
A_2: Keep current strategy	20	0	20

5 Contextualized 4x3 Problem - Technology company

A technology company is deciding which product strategy to adopt for the launch of a new device. The manager knows that market conditions are uncertain and has no reliable information about the probability of each possible level of demand. The company is considering four possible strategies: launching a premium version, launching a standard version, launching a budget version, or delaying the launch.

The level of market demand could be high, medium, or low. The estimated profit for each strategy under each possible market condition is shown in the table below.

What decision should the company make using the Minimax Regret criterion?

Alternative	High demand (S_1)	Medium demand (S_2)	Low demand (S_3)
Premium version (A_1)	80	35	-20
Standard version (A_2)	60	42	18
Budget version (A_3)	40	32	25
Delay launch (A_4)	15	15	15

All payoffs are expressed in thousands of dollars.

Determine the decision using the Minimax Regret criterion.

5.1 Solution

Step 1: Symbols, values, and meanings.

Symbol	Value or identification	Meaning
A_1	Premium version	Alternative chosen before the state is known.
A_2	Standard version	Alternative chosen before the state is known.
A_3	Budget version	Alternative chosen before the state is known.
A_4	Delay launch	Alternative chosen before the state is known.
S_1	High demand	Possible state outside the decision-maker's control.
S_2	Medium demand	Possible state outside the decision-maker's control.
S_3	Low demand	Possible state outside the decision-maker's control.
$P_{ay}(A_1, S_j)$	80, 35, -20 (thousand dollars)	Payoffs for Premium version, in S_1, S_2, S_3 order.
$P_{ay}(A_2, S_j)$	60, 42, 18 (thousand dollars)	Payoffs for Standard version, in S_1, S_2, S_3 order.
$P_{ay}(A_3, S_j)$	40, 32, 25 (thousand dollars)	Payoffs for Budget version, in S_1, S_2, S_3 order.
$P_{ay}(A_4, S_j)$	15, 15, 15 (thousand dollars)	Payoffs for Delay launch, in S_1, S_2, S_3 order.
$b(S_j)$	Calculated in Step 2	Best available payoff in state S_j (column maximum).
$r(A_i, S_j)$	Calculated in Step 3	Opportunity loss: the best payoff in this state minus this alternative's payoff.
$R(A_i)$	Calculated in Step 4	Maximum regret for alternative A_i (row maximum).

All payoff and regret amounts below are in thousands of dollars.

Step 2: Find the best payoff in each state.

Symbolic calculations

State	Comparison
S_1	$\max\{P_{ay}(A_1,S_1), P_{ay}(A_2,S_1), P_{ay}(A_3,S_1), P_{ay}(A_4,S_1)\} = b(S_1)$
S_2	$\max\{P_{ay}(A_1,S_2), P_{ay}(A_2,S_2), P_{ay}(A_3,S_2), P_{ay}(A_4,S_2)\} = b(S_2)$
S_3	$\max\{P_{ay}(A_1,S_3), P_{ay}(A_2,S_3), P_{ay}(A_3,S_3), P_{ay}(A_4,S_3)\} = b(S_3)$

Numerical calculations

State	Comparison
S_1	$\max\{80, 60, 40, 15\} = 80$
S_2	$\max\{35, 42, 32, 15\} = 42$
S_3	$\max\{-20, 18, 25, 15\} = 25$

Step 3: Calculate opportunity losses.

Symbolic calculations

Alternative	S_1	S_2	S_3
A_1	$b(S_1) - P_{ay}(A_1,S_1) = r(A_1,S_1)$	$b(S_2) - P_{ay}(A_1,S_2) = r(A_1,S_2)$	$b(S_3) - P_{ay}(A_1,S_3) = r(A_1,S_3)$
A_2	$b(S_1) - P_{ay}(A_2,S_1) = r(A_2,S_1)$	$b(S_2) - P_{ay}(A_2,S_2) = r(A_2,S_2)$	$b(S_3) - P_{ay}(A_2,S_3) = r(A_2,S_3)$
A_3	$b(S_1) - P_{ay}(A_3,S_1) = r(A_3,S_1)$	$b(S_2) - P_{ay}(A_3,S_2) = r(A_3,S_2)$	$b(S_3) - P_{ay}(A_3,S_3) = r(A_3,S_3)$
A_4	$b(S_1) - P_{ay}(A_4,S_1) = r(A_4,S_1)$	$b(S_2) - P_{ay}(A_4,S_2) = r(A_4,S_2)$	$b(S_3) - P_{ay}(A_4,S_3) = r(A_4,S_3)$

Numerical calculations

Alternative	S_1	S_2	S_3
A_1	$80 - 80 = 0$	$42 - 35 = 7$	$25 - (-20) = 45$
A_2	$80 - 60 = 20$	$42 - 42 = 0$	$25 - 18 = 7$
A_3	$80 - 40 = 40$	$42 - 32 = 10$	$25 - 25 = 0$
A_4	$80 - 15 = 65$	$42 - 15 = 27$	$25 - 15 = 10$

Step 4: Find the maximum regret for each alternative.

Symbolic calculations

Alternative	Row maximum
A_1	$\max\{r(A_1,S_1), r(A_1,S_2), r(A_1,S_3)\} = R(A_1)$
A_2	$\max\{r(A_2,S_1), r(A_2,S_2), r(A_2,S_3)\} = R(A_2)$
A_3	$\max\{r(A_3,S_1), r(A_3,S_2), r(A_3,S_3)\} = R(A_3)$
A_4	$\max\{r(A_4,S_1), r(A_4,S_2), r(A_4,S_3)\} = R(A_4)$

Numerical calculations

Alternative	Row maximum
A_1	$\max\{0, 7, 45\} = 45$
A_2	$\max\{20, 0, 7\} = 20$
A_3	$\max\{40, 10, 0\} = 40$
A_4	$\max\{65, 27, 10\} = 65$

Step 5: Apply the Minimax Regret criterion.

Symbolic calculations

Quantity	Comparison
R^*	$\min\{R(A_1), R(A_2), R(A_3), R(A_4)\} = R^*$
A^*	$\arg\min_i R(A_i) = A^*_{\text{Minimax Regret}}$

Numerical calculations

Quantity	Comparison
R^*	$\min\{45, 20, 40, 65\} = 20$
A^*	$R(A_2) = R^* = 20 = A_2$

Launch the standard version (A_2): it minimizes maximum regret at \$20,000. The other strategies have maximum regrets of \$45,000 (premium), \$40,000 (budget), and \$65,000 (delay). There is no tie.

Step 6: Final numerical opportunity loss table (thousand dollars).

Alternative	S_1	S_2	S_3	Maximum regret
A_1: Premium version	0	7	45	45
A_2: Standard version	20	0	7	**20** (recommended)
A_3: Budget version	40	10	0	40
A_4: Delay launch	65	27	10	65